

Name: .....

Maths Class: .....

# SYDNEY TECHNICAL HIGH SCHOOL



## YEAR 12 HSC COURSE

### Extension 2 Mathematics

#### Assessment 2

June 2014

**TIME ALLOWED: 75 minutes**

#### ***Instructions:***

- ***Start each question on a new page.***
- Write your name and class at the top of this page, and on your answer booklet.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- Write in blue or black pen only.
- It is suggested that you spend no more than 7 minutes on Part A.
- Approved calculators may be used.
- Standard Integrals are supplied at the rear of this paper. This is the only sheet which may be detached from any booklet.



**PART A: (5 Marks)**

Answers to these multiple choice should be completed on the multiple choice answer sheet supplied with your answer booklet.

All questions are worth 1 mark

|   |                                                                                                                                                                                                                                                                                                                                                                    |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>A Primitive of <math>\frac{1}{\sqrt{x^2+16}}</math> is</p> <p>A. <math>\frac{1}{4}\ln(x^2 + 16)</math></p> <p>B. <math>\frac{1}{2}\sqrt{x^2 + 16}</math></p> <p>C. <math>\ln(x + \sqrt{x^2 + 16})</math></p> <p>D. <math>\frac{1}{4}\tan^{-1}(x^2 + 16)</math></p>                                                                                              |
| 2 | <p><math>P(x)</math> is a monic polynomial with real coefficients and has zeros of <math>2 + i</math>, 2 and -2.</p> <p><math>P(x) =</math></p> <p>A. <math>x^4 - 4x^3 + x^2 + 16x - 20</math></p> <p>B. <math>x^4 + 4x^3 + x^2 - 16x - 20</math></p> <p>C. <math>x^4 + 2ix^3 + 9x^2 - 8ix + 20</math></p> <p>D. <math>x^3 - (2 + i)x^2 - 4x + (8 + 4i)</math></p> |
| 3 | <p><math>\int \sec^2 x \tan x dx =</math></p> <p>A. <math>\tan x + k</math>                      B. <math>\tan^2 x + k</math></p> <p>C. <math>\frac{1}{2}\tan^2 x + k</math>                      D. <math>\frac{1}{3}\sec^3 x + k</math></p>                                                                                                                      |

4

If two of its roots are equal, the roots of  $2x^3 + 3x^2 - 12x + 7 = 0$  are:

A.  $x = -1$  and  $x = \frac{7}{2}$

B.  $x = 1$  and  $x = -\frac{7}{2}$

C.  $x = -2$  and  $x = -\frac{7}{8}$

D.  $x = 2$  and  $x = \frac{7}{8}$

5

$$\int \sin^3 x \, dx =$$

A.  $\frac{1}{4}\sin^4 x + k$

B.  $\frac{1}{4}\cos^4 x + k$

C.  $\frac{1}{3}\cos^3 x - \cos x + k$

D.  $-\cos x - \frac{1}{3}\cos^3 x + k$

## PART B

(START EACH QUESTION ON A NEW PAGE)

### QUESTION 6: (10 Marks)

Marks

1 (a) Find  $\int \frac{dx}{1 - \sin^2 x}$

2 (b) Evaluate  $\int_2^7 \frac{x-2}{\sqrt{x+2}} dx$

2 (c) (i) Find values of a, b and c, so that

$$\frac{5x^2 - x - 2}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$

2 (ii) Hence find  $\int \frac{5x^2 - x - 2}{(x+1)(x^2+1)} dx$

3 (d) Sketch the curve,  $y = \ln|\sin x|$  for  $-2\pi \leq x \leq 2\pi$ , showing all keypoints

**QUESTION 7:** (10 Marks) (*Start on a new page*)

**Marks**

- 3 (a) Using t-results, or otherwise, evaluate:

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$$

- 3 (b) The equation  $ax^4 + bx^3 + cx + d = 0$  has a triple root.

Show that  $4a^2c + b^3 = 0$

- 1 (c) (i) Solve the equation  $\cos 5\theta + 1 = 0$  for  $0 \leq \theta \leq 2\pi$

- 3 (ii) Using the substitution  $x = \cos \theta$ ,

solve  $16x^5 - 20x^3 + 5x + 1 = 0$  and hence show that  $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$

(You may assume the result:  $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$ )

**QUESTION 8: (10 Marks)** *(Start on a new page)*

Marks

- 5 (a) By using the substitution  $x = 3\sin \theta$ , or otherwise, show that

$$\int \frac{x^2}{\sqrt{9-x^2}} dx = \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x}{2} \sqrt{9-x^2} + k$$

- (b) The real roots of  $x^3 + 4x - m = 0$  are  $\alpha$ ,  $\beta$  and  $\gamma$ .

1 (i) Find the value of  $\frac{1}{\alpha^2\beta\gamma} + \frac{1}{\alpha\beta^2\gamma} + \frac{1}{\alpha\beta\gamma^2}$

1 (ii) Explain why  $\frac{1}{\alpha^2\beta\gamma} = \frac{1}{m\alpha}$

- 3 (iii) Hence, or otherwise, find the cubic polynomial whose roots are

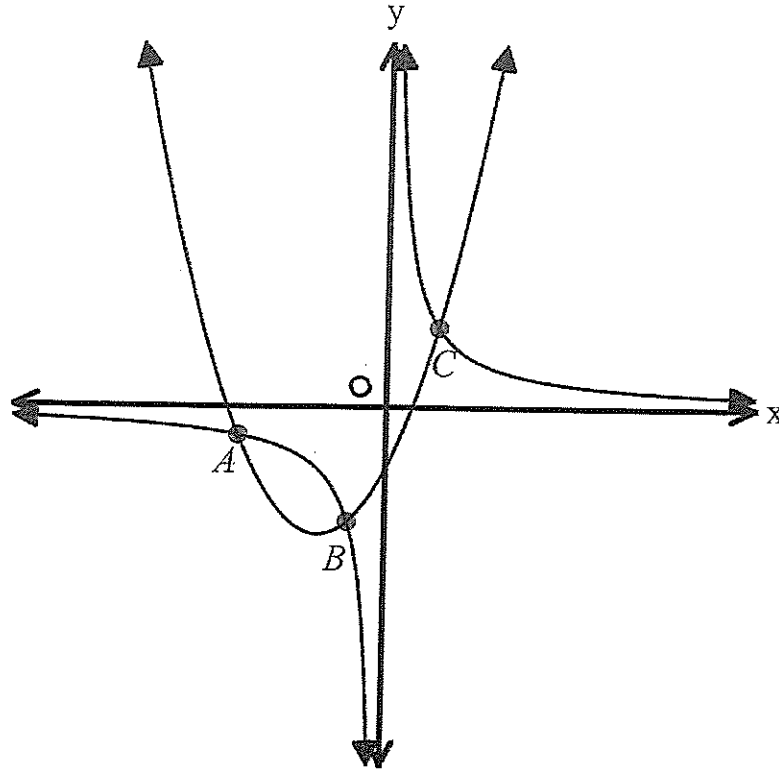
$$\frac{1}{\alpha^2\beta\gamma}, \frac{1}{\alpha\beta^2\gamma}, \text{ and } \frac{1}{\alpha\beta\gamma^2}$$

**QUESTION 9: (10 Marks)** *(Start on a new page)*

Marks

2 (a) Find  $\int \frac{dx}{\sqrt{3+2x-x^2}}$

(b)



In the diagram above, the points A, B and C represent the points of intersection of the curve  $y = x^2 + 2x - 1$  and the curve  $y = \frac{1}{x}$ . O is the origin

The x-values of A, B and C are  $\alpha$ ,  $\beta$ , and  $\gamma$ .

- 1 (i) Show that  $\alpha$ ,  $\beta$ , and  $\gamma$  satisfy  $x^3 + 2x^2 - x - 1 = 0$
- 2 (ii) Find a polynomial with roots  $\alpha^2$ ,  $\beta^2$  and  $\gamma^2$
- 2 (iii) Find the value of  $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$
- 3 (iv) Prove that  $OA^2 + OB^2 + OC^2 = 11$

# SOLUTIONS - YEAR 12 TERM 2

## EXTENSION 2 ASSESSMENT

### Multiple Choice

1/ C      2/ A      3/ C      4/ B      5/ C.

### QUESTION 6:

$$(a) \int \frac{dx}{1 - \sin^2 x} = \int \frac{dx}{\cos^2 x} = \tan x + k$$

$$(b) \int_2^7 \frac{x+2}{\sqrt{x+2}} dx = \int_2^7 \frac{x}{\sqrt{x+2}} dx + \int_2^7 \frac{2}{\sqrt{x+2}} dx$$

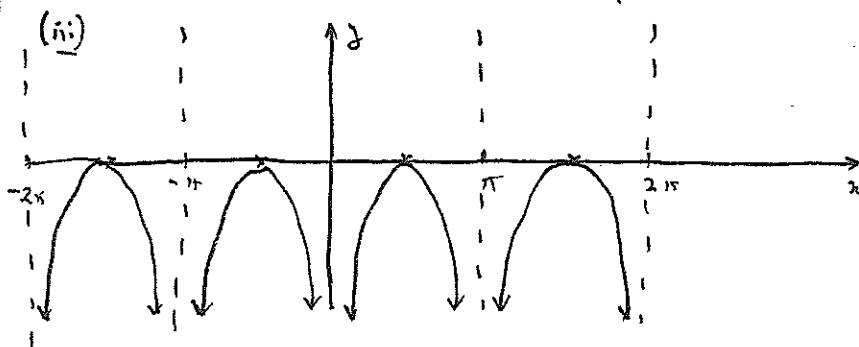
$$= \left[ \frac{2}{3} (x+2)^{3/2} \right]_2^7 - \left[ 8 (x+2)^{1/2} \right]_2^7$$

$$= 18 - \frac{16}{3} - 8(3-2) = \frac{14}{3}$$

(c) (i)  $a = 2, b = 3, c = -4$

$$(ii) \int \frac{5x^2 - x - 2}{(x+1)(x^2+1)} dx = \int \frac{2}{x+1} dx + \int \frac{3x-4}{x^2+1} dx$$

$$= 2 \ln(x+1) + \frac{3}{2} \ln(x^2+1) - 4 \tan^{-1} x + k$$



### QUESTION 7:

$$t = \tan^2 \frac{x}{2} \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$x = 0, t = 0$$

$$x = \frac{\pi}{2}, t = 1$$

$$\therefore dx = \frac{2}{\sec^2 \frac{x}{2}} dt$$

$$= \frac{2}{1+t^2} dt$$

$$\therefore \int_0^1 \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2 dt}{2 + 2t^2 + 1 - t^2}$$

$$= 2 \int_0^1 \frac{dt}{t^2 + 3}$$

$$= 2 \cdot \frac{1}{\sqrt{3}} \left[ \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$(b) \quad P'(x) = 4ax^3 + 3bx^2 + c$$

$$P''(x) = 12ax^2 + 6bx$$

For triple roots,  $P'(x) = 0$

$$\therefore 6x(2ax + b) = 0$$

$$\therefore x = 0 \quad \text{or} \quad x = -b/2a$$

But  $P(x) \neq 0$   $\therefore$  triple root is  $-b/2a$

$$\therefore P'(-b/2a) = 4a(-b/2a)^3 + 3b(-b/2a)^2 + c = 0$$

$$\therefore -\frac{b^3}{2a^2} + \frac{3b^3}{4a^2} + c = 0$$

$$b^3 + 8a^2c = 0$$

(c)(i) Since  $0 \leq \theta \leq 2\pi$ ,  $0 \leq 5\theta \leq 10\pi$ ,

$$\therefore 5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\therefore \theta = \pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5$$

(ii) Let  $x = \cos \theta$

$$\therefore 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta + 1 = 0$$

$$\text{i.e.} \quad \cos 5\theta + 1 = 0$$

$$\therefore \theta = \pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5 \quad (\text{from above})$$

$$\therefore x = \cos \theta = \cos \pi/5, \cos 3\pi/5, \cos \pi, \cos 7\pi/5, \cos 9\pi/5$$

$$\text{and} \quad \cos 7\pi/5 = \cos 3\pi/5, \quad \cos 9\pi/5 = \cos \pi/5$$

$\therefore$  By sum of roots,

$$\cos \pi/5 + \cos 3\pi/5 - 1 + \cos 3\pi/5 + \cos \pi/5 = 0$$

$$\therefore 2(\cos \pi/5 + \cos 3\pi/5) = 1$$

$$\therefore \cos \pi/5 + \cos 3\pi/5 = +1/2$$

QUESTION 8:

$$(a) \quad x = 3 \sin \theta \Rightarrow \frac{dx}{d\theta} = 3 \cos \theta$$

$$dx = 3 \cos \theta d\theta$$

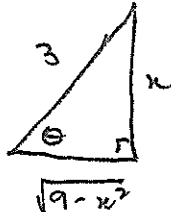
$$\therefore \int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 \theta}{\sqrt{9-9 \sin^2 \theta}} \cdot 3 \cos \theta d\theta$$

$$= \int 9 \sin^2 \theta d\theta$$

$$= \frac{9}{2} \int 1 - \cos 2\theta d\theta$$

$$= \frac{9}{2} \theta - \frac{9}{4} \sin 2\theta + k$$

$$= \frac{9}{2} \theta - \frac{9}{2} \sin \theta \cos \theta + k$$

Now  (from  $x = 3 \sin \theta$ )  
gives  $\cos \theta = \frac{\sqrt{9-x^2}}{3}$

$$\therefore \text{Integral} = \frac{1}{2} \sin^{-1} \frac{x}{3} - \frac{1}{2} \cdot \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3} + h$$

$$= \frac{1}{2} \sin^{-1} \frac{x}{3} - \frac{x}{6} \sqrt{9-x^2} + h$$

(b) (i) 
$$\begin{cases} \alpha + \beta + \gamma = 0 \\ \alpha\beta + \alpha\gamma + \beta\gamma = 4 \\ \alpha\beta\gamma = m \end{cases}$$

$$\therefore \frac{1}{\alpha^2\beta\gamma} + \frac{1}{\alpha\beta^2\gamma} + \frac{1}{\alpha\beta\gamma^2} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{4}{m^2}$$

(ii) Since  $\alpha\beta\gamma = m$   
then  $\alpha^2\beta\gamma = \alpha m$   
 $\Rightarrow \frac{1}{\alpha^2\beta\gamma} = \frac{1}{\alpha m}$

(iii) Polynomial is  $x^3 - (\text{sum of roots})x^2 + (\text{sum in pairs})x - \text{product} =$

$$\text{Sum in pairs} = \frac{1}{\alpha\beta^2\gamma^2} + \frac{1}{\alpha^2\beta^3\gamma^2} + \frac{1}{\alpha^2\beta^2\gamma^3}$$

$$= \frac{\alpha + \beta + \gamma}{\alpha^3\beta^3\gamma^3} = 0 \text{ (from above)}$$

$$\text{Product} = \frac{1}{\alpha^4\beta^4\gamma^4}$$

$$= \frac{1}{m^4}$$

$\therefore$  Polynomial is  $x^3 - \frac{4}{m^2}x^2 + 0x - \frac{1}{m^4} = 0$  ← from part (i)

$$\therefore m^4 x^3 - 4m^2 x^2 - 1 = 0$$

QUESTION 9:

(a) 
$$\int \frac{dx}{\sqrt{3+2x-x^2}} = \int \frac{dx}{\sqrt{3-(x-1)^2+1}}$$

$$= \int \frac{dx}{\sqrt{4-(x-1)^2}}$$

$$= \sin^{-1} \left( \frac{x-1}{2} \right) + h$$

(b) (i) Intersection of  $y = \frac{1}{x}$  and  $y = x^3 - 2x - 1$

$$\text{ie } x^3 - 2x - 1 = \frac{1}{x}$$

$$\therefore x^4 - 2x^2 - x - 1 = 0$$

$$\therefore x^4 - 2x^2 - x - 1 = 0$$

(ii)  $P(\sqrt{x}) = (\sqrt{x})^3 + 2(\sqrt{x})^2 - (\sqrt{x}) - 1 = 0$

$$\sqrt{x}(x-1) = 1 - 2x$$

$$\therefore x(x-1)^2 = (1-2x)^2$$

$$\therefore x^3 - 2x^2 + x = 1 - 4x + 4x^2$$

$$\therefore x^3 - 4x^2 + 5x - 1 = 0$$

$$= 0$$

(iii) In above,  $\alpha^2 + \beta^2 + \gamma^2 = 6$

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = 5$$

$$\alpha^2\beta^2\gamma^2 = 1$$

$$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha^2\beta^2\gamma^2}$$

$$= 5.$$

(iv) If M is the foot of the perpendicular from A to the x-axis.

$$OA^2 = \alpha^2 + \left(\frac{1}{\alpha}\right)^2 \text{ by distance formula.}$$

$$\therefore OA^2 + OB^2 + OC^2 = \alpha^2 + \left(\frac{1}{\alpha}\right)^2 + \beta^2 + \left(\frac{1}{\beta}\right)^2 + \gamma^2 + \left(\frac{1}{\gamma}\right)^2$$

$$= \alpha^2 + \beta^2 + \gamma^2 + \left(\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}\right)$$

$$= 6 + 5$$

$$= 11$$